

Thinking Beyond Images: Using Language to Improve Image Classification

Javon Hickmon, Sarah Pratt, Ali Farhadi

1. Introduction

- There have been many significant improvements in Multi-Modal learning
 - Multi-Modal Learning** aims to train models that learn patterns between different forms of data
 - Flamingo** by DeepMind
- Recent efforts have open-sourced certain Multi-Modal model architectures
- Chain-of-Thought** prompting has been shown to improve performance on a range of tasks^[3]
 - Chain-of-Thought is a series of intermediate reasoning steps^[3]

Our question: Do the benefits from Chain-of-Thought carry over to the Multi-Modal domain?

2. Methods

- Flamingo:** A Vision Language Model for Few Shot Learning
 - Bridges powerful vision-only and language-only models^[1]
- CuPL** (Customized Prompts via Language models)
 - Leverage the knowledge contained within Large Language Models to improve zero-shot image classification^[2]
- Chain-of-Thought**
 - Significantly improves the ability of large language models to perform complex reasoning^[3]

3. Next Steps

- Analyze** common mistakes, and adjust the prompting structure to reduce inaccuracies
- Quantify** the accuracy of the model on a range of image classification benchmarks
- Observe** whether or not the model can still achieve state-of-the-art performance^[3]
 - The literature shows this is highly probable

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

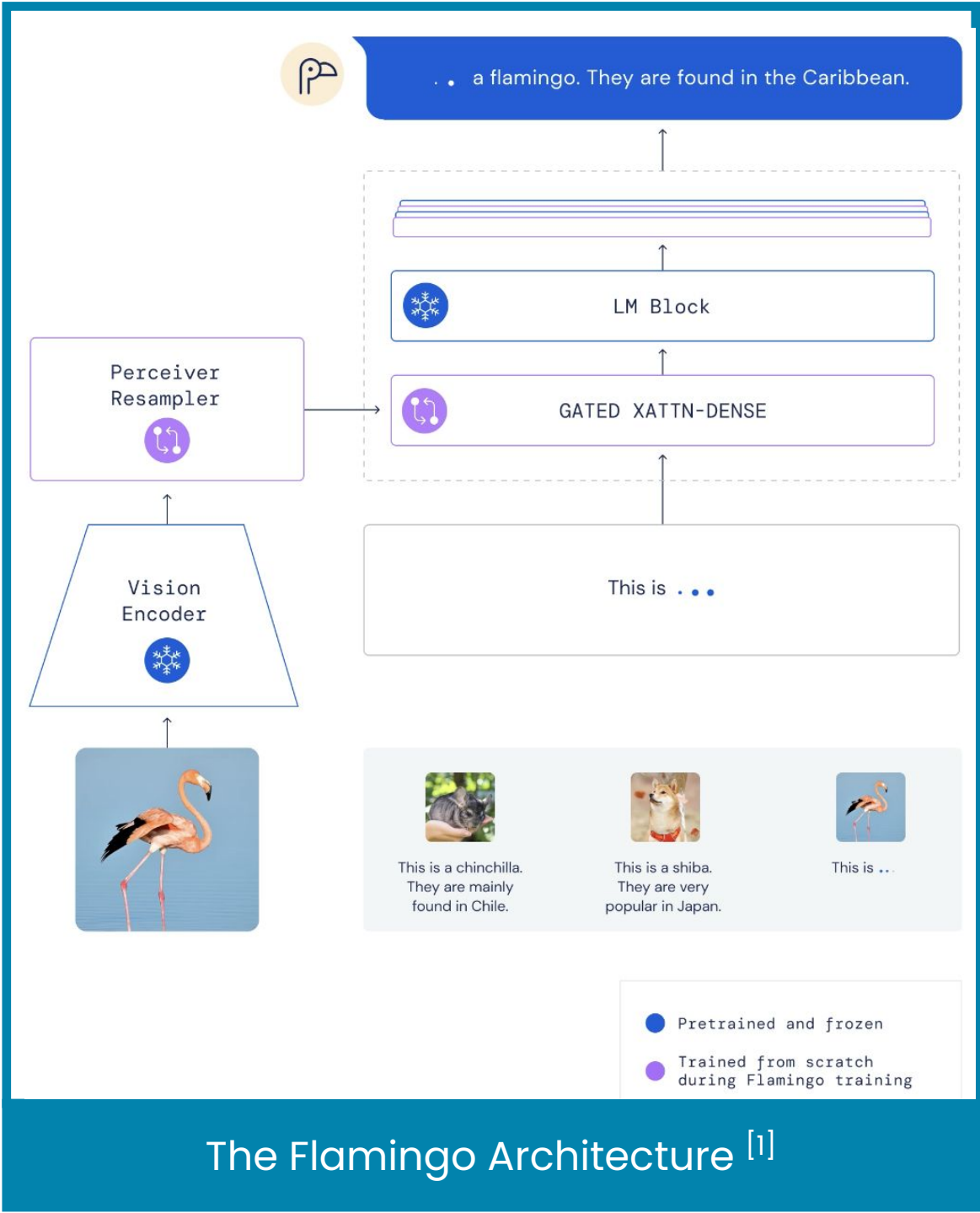
A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✓

Example Chain-of-Thought Prompt ^[3]



We aim to show...

Chain-of-Thought improves classification accuracy for Multi-Modal Models



- What is in this picture?
- There is a canine with short golden and white fur and pointy ears.
- Therefore, this is a Welsh Corgi.



- What is in this picture?
- There is a duck-billed mammal with brown fur.
- Therefore, this is a platypus.



References

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- [2] Pratt, S., Liu, R., & Farhadi, A. (2022). What does a platypus look like? Generating customized prompts for zero-shot image classification. *arXiv preprint arXiv:2209.03320*.
- [3] Wei, J., Wang, X., Schuurmans, D., Bosma, M., Chi, E., Le, Q., & Zhou, D. (2022). Chain of thought prompting elicits reasoning in large language models. *arXiv preprint arXiv:2201.11903*.